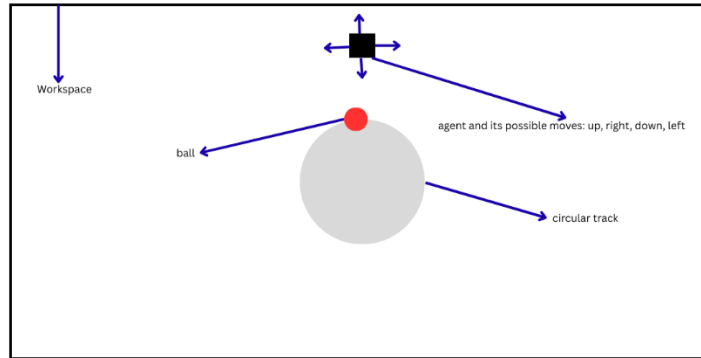


## The Virtual PULSR Environment

1. Description: A simple 2D path tracking environment where an agent (black box) follows the red ball through the boundaries of the circle in a complete revolution. The goal is to find the shortest trajectory of a complete revolution while maintaining close proximity to the red ball.
2. Physical Outlook



3. State Space: x coordinate, y coordinate and score {score  $\pm 1$  if  $d \leq 70$ , where  $d$  is the Euclidean distance between the agent and the ball}.
4. Observation Space: Same as State Space.
5. Action Space: Up, Right, Down, Left ("N", "E", "S", "W"). If the agent hits the workspace boundary, it will reverse the initial step and find a new path.
6. Reward Function:

| Final Score of the Episode | Reward |
|----------------------------|--------|
| >150                       | 3      |
| 100 – 150                  | 2      |
| 90 – 100                   | 1      |
| 90                         | 0      |
| 70 – 90                    | -1     |
| 50 – 70                    | -2     |
| <50                        | -3     |

Reward Per Step =  $-3 / 239$  approximately -0.01255 (negative because it is trajectory optimization task)

7. Environment Dynamics: If  $d \geq a$ , the agent chooses actions randomly from the action space until  $d \leq a$ . Choose  $a = 50$  for start but change, if need be, to optimize score.